

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I V. Deepak-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

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29/12/20

Annexure A

Smart Campus Network Subnet Allocation
Given;

Network Address: 192.168.100.0/24

School of Computing: 125

School of Electronics: 126

School of Mechanical Engineering: 127

Solution;

School of Computing (125):

Network address: 192.168.100.0

Host Range: 192.168.100.1 - 192.168.100.126

Broadcast Address: 192.168.100.127

Usable Hosts: 126,

School of Electronics (126):

Network address: 192.168.100.128

Host Range: 192.168.100.129 - 192.168.100.190

Broadcast Address: 192.168.100.191.

Usable Hosts: 62.

School of Mechanical Engineering (127) Quality Assurance

Network Address: 192.168.100.192

Host Range: 192.168.100.193 - 192.168.100.223

Broadcast Address: 192.168.100.223

Usable Hosts: 30

2) Department-wise Subnet Allocation for a Software Company.

Given,

Network Address: 10.10.0.0/16

Software Development $\rightarrow$ /24

Quality Assurance $\rightarrow$ /25

Human Resources $\rightarrow$ /26

Executive Management $\rightarrow$ /27

Solution,

• Software Development (/24):

$\rightarrow$ Network Address: 10.10.0.0

$\rightarrow$ Host Range: 10.10.0.1 - 10.10.0.254

$\rightarrow$ Broadcast Address: 10.10.0.255

$\rightarrow$ Usable Hosts: 254

Quality Assurance (125):

- > Network Address: 10.10.1.0
- > Host Range: 10.10.1.1 - 10.10.1.126
- > Broadcast Address: 10.10.1.127
- > Usable Hosts: 126

• Human Resources (126):

- > Network Address: 10.10.1.128
- > Host Range: 10.10.1.129 - 10.10.1.190
- > Broadcast Address: 10.10.1.191
- > Usable Hosts: 62

• Executive Management (127):

- > Network Address: 10.10.1.192
- > Host Range: 10.10.1.193 - 10.10.1.222
- > Broadcast Address: 10.10.1.223
- > Usable Hosts: 30

3) IPv6 Address Migration for a Smart Hospital

Green,

IPv6: 2001:0db8:abcd:0000:6000:0000:0000:0000/64

Solution,

Compressed IPv6 address:

• 2001:db8:abcd::164

Expanded: 2001:0db8:abcd:0000:0000:0000:0000:0000

Network prefix:

→ 2001:db8:abcd:0::/64

Total possible Host Addresses:

→ $2^{64} = 18,446,744,073,709,551,616$.

The IPv6 is successfully compressed and expanded. Using a /64 prefix, the subnet supports 2^{64} possible host addresses, making it suitable for large scale IoT deployment.

4) Routing Table Lookup and Packet Forwarding.

Given,

Routing Table:

→ 192.168.1.0/24

→ 192.168.2.0/24

→ 192.168.3.0/24

Procedure:

Compare the destination IP with routing table

entries to identify the matching Subnet and determine forwarding path.

Solution,

Destination Subnet:

$\rightarrow 192.168.2.0/64$

Matching Routing Entry:

→ 192.168.2.0/24

Next - 1 top Network Interface:

-) Interface connected to 192.168.2.0/24 network.

5) LAN Configuration for Two Office Departments

Criven,

- Administration LAN $\rightarrow 192.168.10.0/26$
- Finance LAN $\rightarrow 192.168.10.64/26$
- Administration Gateway $\rightarrow 192.168.10.1$
- Finance Gateway $\rightarrow 192.168.10.65$

Procedure,

Determine the network details for each LAN, assign IP addresses to three computers, and identify default gateway.

Solution,

• Administration LAN:

→ Network Address: 192.168.10.0

→ Host Range: 192.168.10.1 - 192.168.10.62

→ Broadcast Address: 192.168.10.63

→ Computer IPs: 192.168.10.2, 192.168.10.3, 192.168.10.4

→ Default Gateway: 192.168.10.1

• Finance LAN:

→ Network Address: 192.168.10.64

→ Host Range: 192.168.10.65 - 192.168.10.126

→ Broadcast Address: 192.168.10.66, 192.168.10.67,

→ Computer IPs: 192.168.10.68, 192.168.10.69, 192.168.10.70

→ Default Gateway: 192.168.10.65

Done ✓